

[cmp653 > final > defense]

Privacy-Budget Consumption in Repeated Aggregate SQL Workloads

> a statistical model _

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> 01 ::: aggregate queries are not safe

// intuition (wrong)

“aggregates return totals, not rows
- safe.”

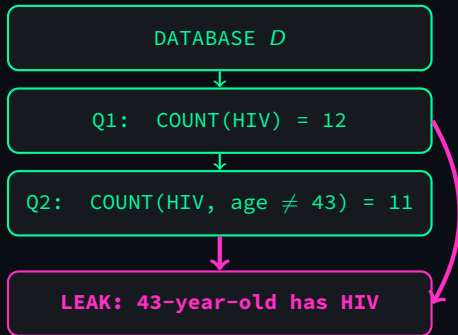
// reality

two overlapping COUNTs isolate one
person.

```
- 12 patients with HIV  
SELECT COUNT(*) FROM patients  
WHERE zip='06897' AND  
      disease='HIV';
```

```
- 11 with HIV, age <> 43  
SELECT COUNT(*) FROM patients  
WHERE zip='06897' AND  
      disease='HIV' AND  
      age <> 43;
```

⇒ the 43-year-old in 06897 has HIV.



Dinur & Nissim 2003:

$O(n)$ noisy aggregates reconstruct the
whole DB.

> 02 ::: differential privacy in 30 seconds

ϵ -differential privacy

$\mathcal{M}$ is ϵ -DP if for any $D \sim D'$ (differ by 1 row):

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(D') \in S]$$

// Laplace mechanism

$$\tilde{f}(D) = f(D) + \eta, \quad \eta \sim \text{Lap}(\Delta f / \epsilon)$$

// budget composes additively across queries.

// example: $Q = \text{COUNT}(\ast) = 1000$

ϵ	scale	answer
1.0	± 1	999-1001
0.1	± 10	990-1010
0.01	± 100	900-1100

smaller $\epsilon \Rightarrow$ stronger privacy + more noise

the engineering question

A finite total budget ϵ_{total} is consumed as the workload runs.

HOW does it degrade? Predicting that before deployment is what we do.

> 03 ::: milestone version (what i had)

- ▶ Python middleware: SQL parser + sensitivity analyzer + Laplace mechanism
- ▶ Budget ledger with exact-repeat caching
- ▶ TPC-H benchmark, 4 workload families (W1-W4)
- ▶ 28 unit tests, working prototype

milestone headline

“Workload-aware caching saves 100× budget on a repetitive workload by exploiting DP’s post-processing property.”

// i thought this was the contribution.

> 04 :: based on the review i received

// points the reviewer raised:

- ▶ caching's role in DP wasn't clear
- ▶ Algorithm 1 was too thin
- ▶ $1 - 1/k$ left unexplained
- ▶ AVG mechanism wasn't justified
- ▶ caching as a contribution is debatable

// and a structural ask:

“exact-repeat assumptions are strong.

budget and temporality should be considered together.

leakage / savings models are limited.

MECHANISM

(caching)

= **debatable**



MODEL

(workload structure $\rightarrow$ budget)

= **contribution**

// the framing of the contribution
had to flip.

> 05 :: research question, reframed

old: “does caching help?” // trivial

new:

Given a workload distribution and a privacy budget, can a closed-form statistical model predict the realized budget consumption and per-query utility before any query is issued?

Caching is now the corner case of the model, not the contribution.

// five concrete deliverables:

1. **Analytical model** (workload structure $\rightarrow$ budget)
2. **Temporal extension** (staleness + updates)
3. **Predictive allocator** (new mechanism)
4. **Leakage analysis** (MIA + reconstruction)
5. **Empirical campaign** (model validation)

> 06 ::: the model in one slide

// setup: k queries drawn i.i.d. from $\{p_i\}_{i=1}^m$.

prop 1 - expected unique queries

$$\mathbb{E}[u_k] = \sum_{i=1}^m [1 - (1 - p_i)^k]$$

prop 2 - workload-aware budget

$$\mathbb{E}[\varepsilon_{wa}] = \varepsilon_q \cdot \mathbb{E}[u_k]$$

two limits:

► $\alpha \rightarrow \infty$: $S(k) = 1 - 1/k$
milestone's toy formula

► $\alpha \rightarrow 0$: $S(k) \rightarrow 0$

// concrete example: dashboard
with 5 tiles

tile	prob p_i	meaning
A	0.42	"urgent"
B	0.26	"high"
C	0.17	"medium"
D	0.10	"low"
E	0.05	"other"

Zipf($\alpha = 1$) over $m = 5$

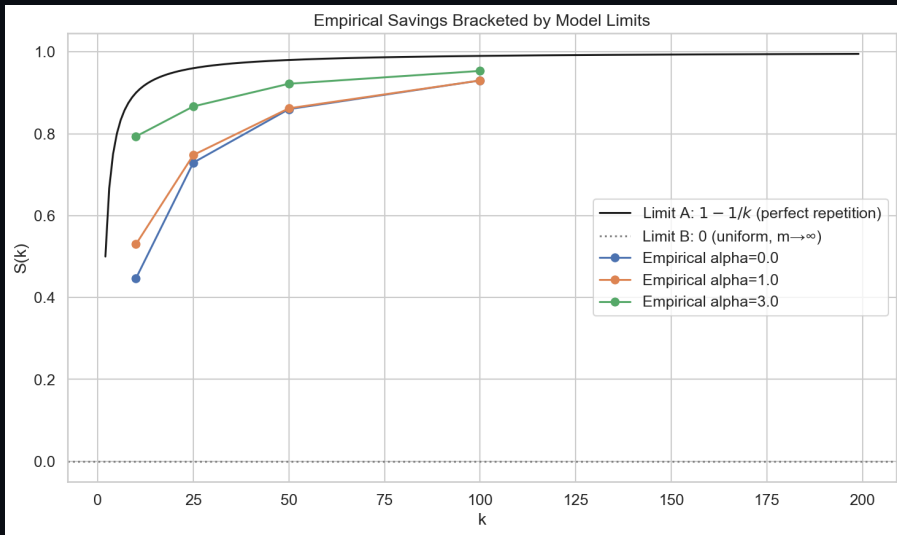
run $k = 100$ queries:

prop 1 predicts $\mathbb{E}[u_k] \approx 4.97$

prop 2 predicts budget
 $= \varepsilon_q \cdot 4.97$

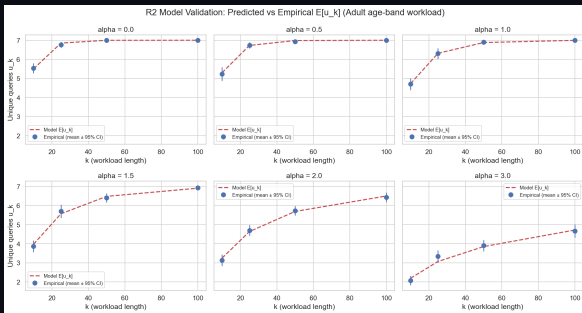
$\Rightarrow$ 95% savings vs naive's $100\varepsilon_q$

> 07 ::: the two limits, empirically



dashed: model prediction | markers: empirical mean over 30 trials

> 08 ::: main grid – 21 of 24 cells within 3%



// markers should land on the dashed identity

// setup

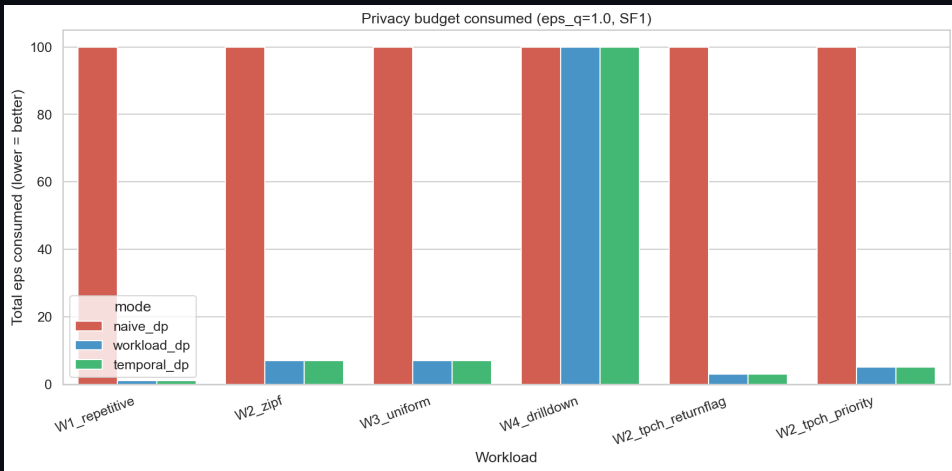
- ▶ $\alpha \in \{0, 0.5, 1, 1.5, 2, 3\}$
- ▶ $k \in \{10, 25, 50, 100\}$
- ▶ 30 trials per cell
- ▶ → 720 trials

// headline

- ▶ 21 / 24 cells within 3%
- ▶ Worst case 8.6% at small k , high α

Predict privacy cost from the workload distribution – before any query runs.

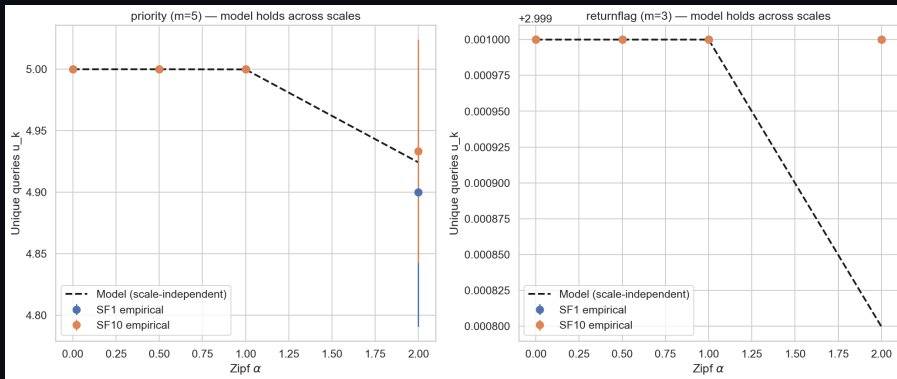
> 09 :: full benchmark – 6 workloads x 3 modes



$\epsilon_q = 1$, total budget $B = 100$, $k = 100$, 30 trials per cell.

red = naive (consumes full B) | blue = workload-DP | green = temporal-DP

> 10 ::: scale-independent (sf=1 vs sf=10)



TPC-H SF=1 (6M rows) vs SF=10 (60M rows)

Empirical $\mathbb{E}[u_k]$ identical within 0.03 units

$\Rightarrow$ the model captures workload structure, not data scale.

> 11 ::: heterogeneous workload – exact match

// W1 + W4 hybrid: fraction f repetitive, $1-f$ unique.
by construction, true $u_k = 1 + (1-f) \cdot k$.

f	true u_k	empirical ε	predicted $\varepsilon_q \cdot u_k$
0.1	91	45.5	45.5
0.3	71	35.5	35.5
0.5	51	25.5	25.5
0.7	31	15.5	15.5
0.9	11	5.5	5.5

⇒ empirical matches prediction to the cent in every cell.

// closed form $\mathbb{E}[\varepsilon_{wa}] = \varepsilon_q \cdot u_k$ holds without requiring Zipf parametricity.

> 12 ::: predictive allocator – the new mechanism

// use the model's $\mathbb{E}[u_k]$ estimate at runtime to choose ε_q :

1. observe template frequencies online (first 3-10 queries: warmup)
2. estimate $\hat{U} = \mathbb{E}[u_k \mid \hat{p}]$ from Proposition 1
3. allocate $\varepsilon_q^{\text{pred}} = B/\hat{U}$ (matches the offline optimum)

// example: $B = 10$, $k = 100$. Naive picks $\varepsilon_q = 10/100 = 0.1$. If we observe $\hat{U} = 7$ unique templates, predictive picks $\varepsilon_q = 10/7 \approx 1.43$ – 14× tighter noise per release.

novel because

PINQ / PrivateSQL / Chorus / DOP-SQL all fix ε_q offline.
This is the first DP-SQL mechanism that adapts ε_q at runtime from a learned model.

B	Mode	MAE	vs naive
1	naive	101.1	–
	predictive	68.7	–32%
10	naive	10.1	–
	predictive	8.3	–18%

> 13a ::: semantic cache – the idea

// step 1: SQL → AST

```
SELECT COUNT(*)  
FROM patients  
WHERE age > 30
```

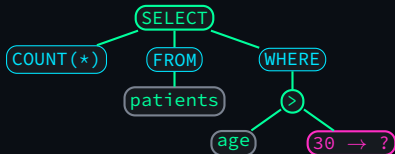
// step 2: normalize literals
→ ?

```
SELECT COUNT(*)  
FROM patients  
WHERE age > ?
```

// step 3: compare two ASTs

- Tree Kernel: count shared subtrees
- AST embed: cosine on 384-dim vec

// step 1 visualized (AST tree):



// literal '30' becomes a placeholder so
two queries with different ages share the same AST.

> 13b ::: semantic cache – honest negative

false positive (admits wrong match)

Q1: SELECT COUNT(*) FROM adult WHERE
age $\geq$ 20 AND age < 30

Q2: SELECT COUNT(*) FROM adult WHERE
age $\geq$ 40 AND age < 50

Same AST template after literal removal.

Very different true counts!

$\Rightarrow$ cache returns wrong answer by 1000s

false negative (misses true match)

Q1: ... WHERE a AND b

Q2: ... WHERE b AND a

Tree Kernel uses ordered children:

kernel score < 0.95 $\Rightarrow$ miss

takeaway

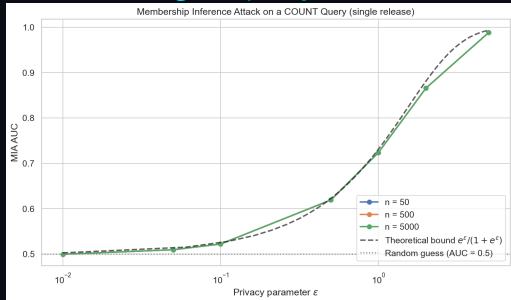
semantic similarity $\neq$ formal equivalence

- ▶ AI tools can measure how alike two queries look
- ▶ but DP requires provable equivalence of answers
- ▶ “AI-powered DP cache” alone is not DP-safe

Future direction: pair similarity tools with a symbolic equivalence prover (predicate normalization, range merging, alpha-conversion) before admitting any cache match.

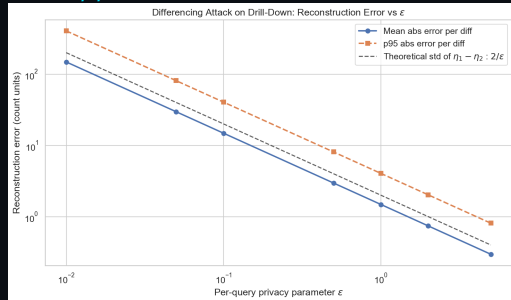
> 14 ::: privacy leakage, quantified

// single-query MIA AUC



empirical tracks theoretical
 $e^\epsilon / (1 + e^\epsilon)$ within 1%

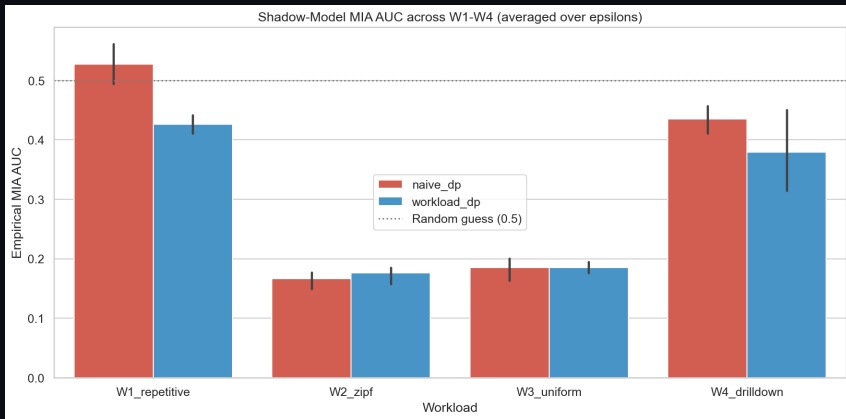
// reconstruction error



differencing on 5-level
drill-down, error $\sim \sqrt{2}/\epsilon$

⇒ implementation matches the theory in the tight regimes.

> 15 ::: shadow MIA across W1-W4



logistic-regression classifier trained on simulated middleware outputs
AUC stays at 0.15-0.58 even though the cumulative- ϵ bound approaches 1.0
 $\Rightarrow$ cumulative- ϵ is loose for realistic workloads.

> 16 ::: what we found – four takeaways

1. Workload-aware budget is structural, not a cache trick.

$\mathbb{E}[\varepsilon_{\text{wa}}] = \varepsilon_q \cdot \mathbb{E}[u_k]$ predicts within 3% on the main grid, **exact** on the mixed workload, same at SF=1 and SF=10.

2. The $1 - 1/k$ toy rule is the $\alpha \rightarrow \infty$ corner case.

The model recovers it **and** the entire interior up to naive composition.

3. Cumulative- ε overestimates real attack success.

Shadow MIA on W1-W4 stays at AUC 0.15–0.58 vs. a 1.0 cumulative bound – budget is over-provisioned in practice.

4. Semantic similarity $\neq$ equivalence.

Tree-kernel + embedding admits false positives **and** misses textbook equivalences. “AI-powered DP” alone is not DP-safe.

> 17 ::: limitations + future work

// what this is not:

- ▶ Not a head-to-head reimplementation of PINQ / PrivateSQL / Chorus / DOP-SQL
- ▶ Not validated on a real workload trace (Zipf is a controlled sweep)
- ▶ Not a new cryptographic privacy mechanism

// concrete next steps:

- ▶ Cache re-release on budget surplus
- ▶ Rényi DP / zCDP composition
- ▶ Burstiness via renewal processes
- ▶ Real workload traces
- ▶ Joint budget-leakage bound
- ▶ Equivalence-checked semantic cache
- ▶ Multi-table / JOIN via residual sensitivity

> **thank you _**

questions ?

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repo: github.com/canoztas/cmp653-project